

PROFESSIONAL PORTFOLIO

This portfolio highlights the academic journey, experiences, projects, achievements, and skills of Siti Asih Rahmahaya, a Geophysical Engineering Fresh graduate from Institut Teknologi Bandung.

SITI ASIH RAHMAHAYA

Geophysical Engineering Fresh Graduate – Institut Teknologi Bandung

TABLE OF CONTENTS

ABOUT ME

KEY SKILLS

EDUCATION

PROJECT EXPERIENCES

ORGANIZATION & LEADERSHIP

ACHIEVEMENTS

ACTIVITIES

CONTACT

Siti Asih Rahmahaya, or Asih, is a Fresh Graduate in **Geophysical Engineering** from **Institut Teknologi Bandung (ITB)** with experience in **geophysical surveys, seismic data acquisition, processing, and interpretation**. She has **strong interests** in **seismic imaging, subsurface mapping, and energy exploration**, and has developed **solid analytical, communication, and teamwork skills** through academic and organizational experiences. **Motivated, detail-oriented, and adaptable**, Asih is eager to apply her technical and problem-solving abilities in **geoscience** and **energy-related fields**.



About Asih

KEY SKILLS

Geoscience Software:



Project Officer:



Programming:



Survey & Acquisition : Gravity Data Acquisition & Processing, Magnetic Data Acquisition & Processing, Geolistrik (Sounding & Profiling Method) Data Acquisition & Processing, ADMT Data Acquisition & Processing, Electromagnetic Data Acquisition & Processing, Seismic Refraction Data Acquisition & Processing, HVSR Data Acquisition Data & Processing, Field Project Survey.

Interpretation & Processing Seismic Data : Fault & Horizon Picking, Seismic Attributes, AVO Analysis, Seismic Inversion, Geological & Subsurface Mapping, Seismic Data Imaging, Spectral Decomposition, Seismic Data Filtering, Seismic Data Enhancing, Denoising & Deconvolution, Trim Static (Pre-Conditioning Seismic Data).

Software : Petrel · Hampson-Russell Suite · Paleoscan · QGIS · ArcGIS · Google Earth · Surfer · Model Vision · Oasis Montaj · IP2WIN · Res2dinv · AIDU Exploration · EM38MK2xp.exe · Refrapy · Seisfra · Geopsy · Global Mapper · ProMax · Viki · Microsoft Office (Word, Excel, PowerPoint) · Canva · Corel Studio · Notion

Programming : Python, MATLAB

Language: English Proficiency (Advanced – spoken, listen & written), TOEFL ITP Score: 543

Soft Skills : Project Management · Teamwork · Adaptability · Leadership · Analytical Thinking Problem Solving · Communication

EDUCATION

Institut Teknologi Bandung (2021–2025)

Geophysical Engineering

Thesis: Reservoir Characterization Of Menggala Formation Sandstone Using A Trigonometric Approach to Pseudo Elastic Impedance Lithology (PEI-L) And Curved Pseudo Elastic Impedance (CPEI) Attributes In The “YUNORA” Field, Central Sumatra Basin.

SMA Negeri 1 Sukaresmi (2018–2021)

Activities and societies: Debate club, Science Club, Entrepreneur and Hydroponics Club.

- General Secretary of Science Club KIR-IPA SMANLA 2020–2021
- Honorable Mention of Earth Science National Olympiad 2020
- 2nd place in Indonesian debate competition (LDBI) regional level 2019
- 2nd place in Indonesian debate competition (CGUD) provincial level 2020

THESIS PROJECT

Thesis Tittle: Reservoir Characterization Of Menggala Formation Sandstone Using A Trigonometric Approach to Pseudo Elastic Impedance Lithology (PEI-L) And Curved Pseudo Elastic Impedance (CPEI) Attributes In The “YUNORA” Field, Central Sumatra Basin.

Well to Seismic Tie

Pre-Conditioning Data

AVO Analysis

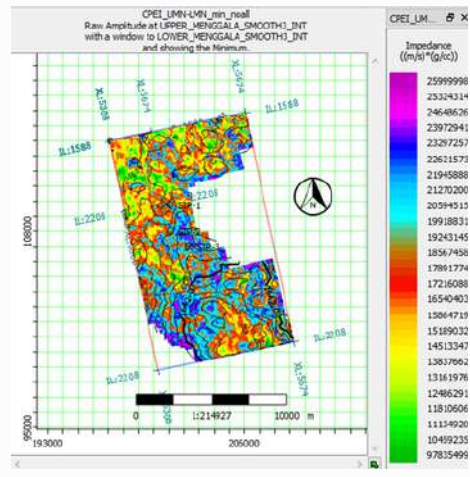
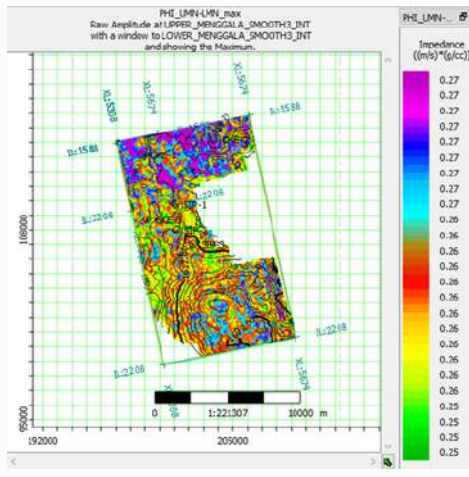
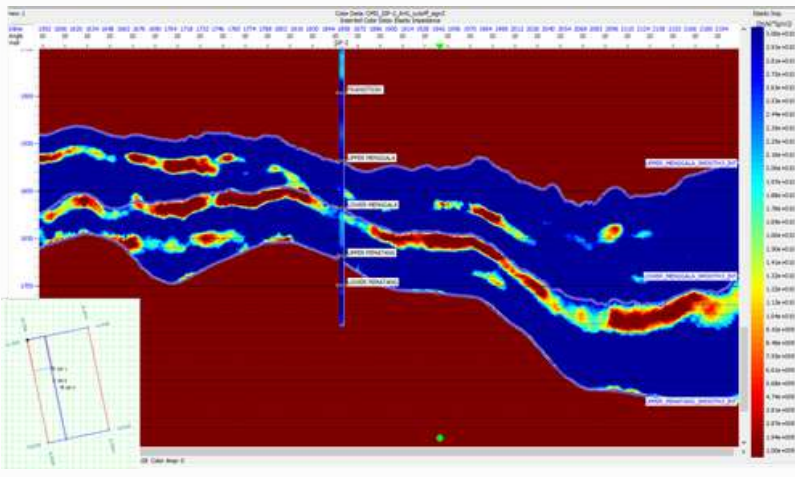
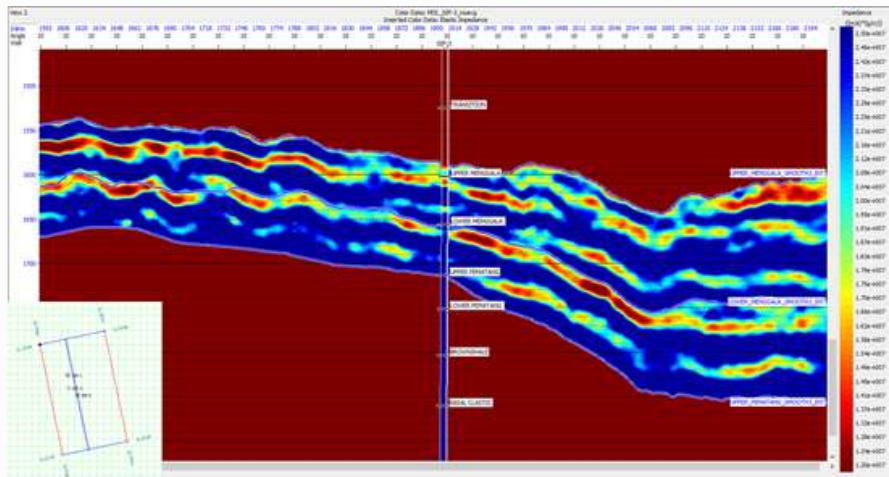
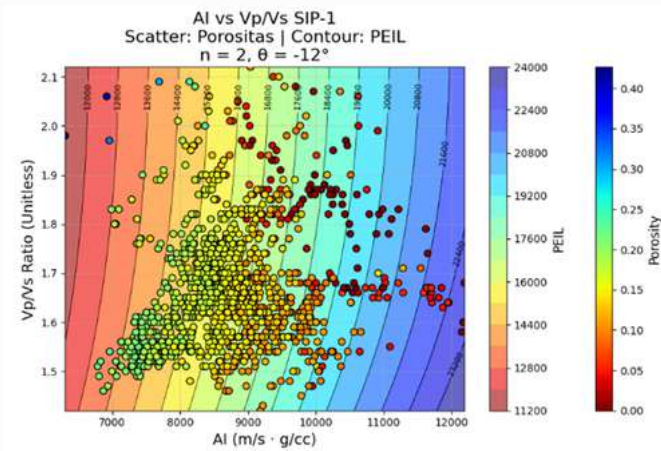
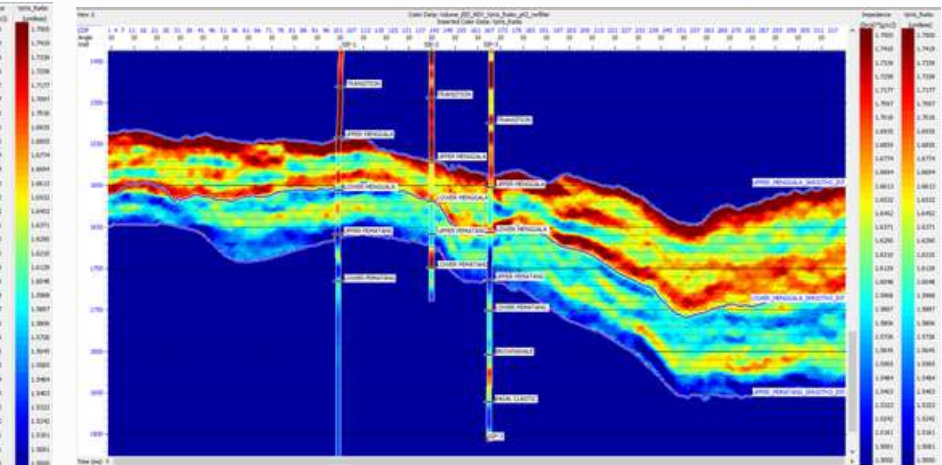
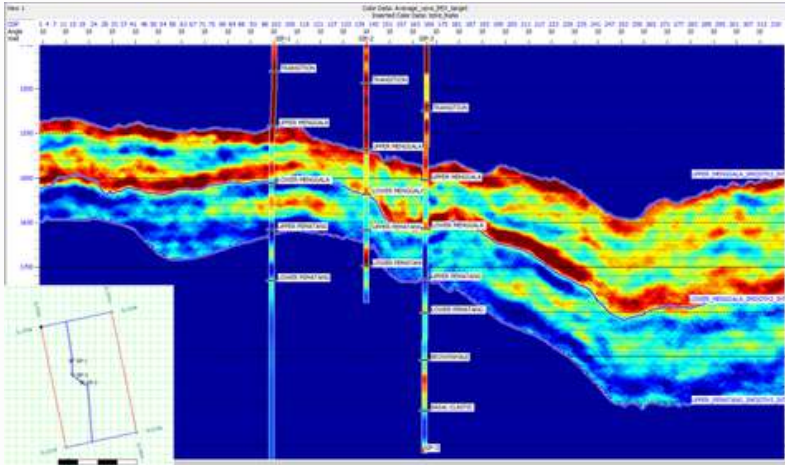
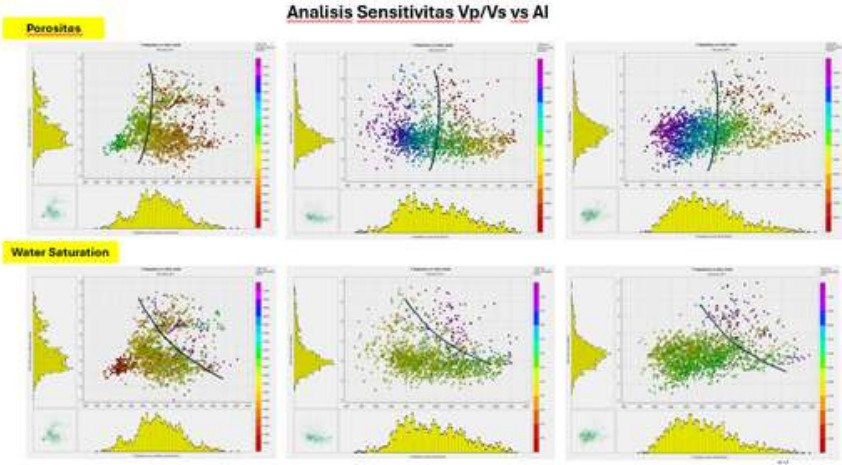
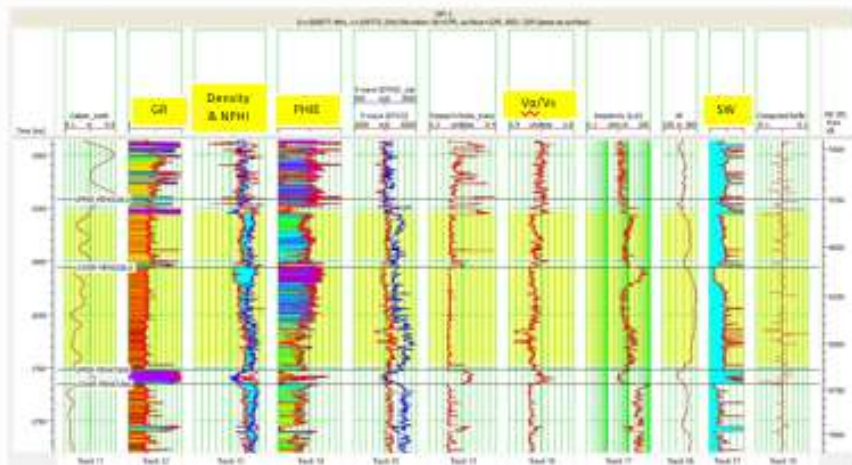
Horizon & Fault Picking

Reservoir Characterization

Seismic Inversion & Attributes

CPEI & PEIL Attributes

Seismic Stratigraphy Analysis



Professional Experience

PT Saka Energi Indonesia (Subsidiary of PGN – Pertamina Gas Negara)
Subsurface Geophysics – Practical Work Program (internship)

Situation

Conducted seismic data interpretation: horizon picking, depth surface mapping, attribute analysis, and GRV calculation.

Task

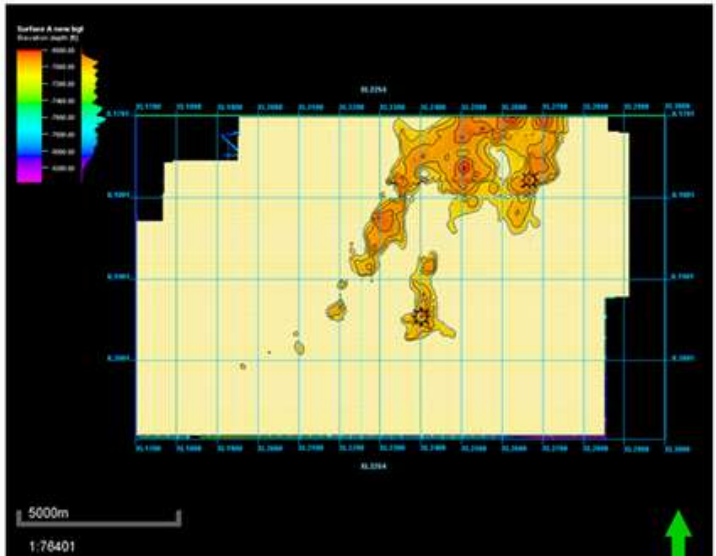
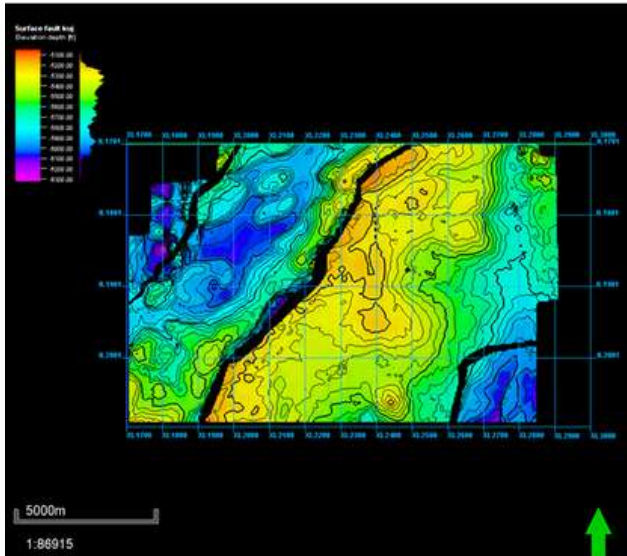
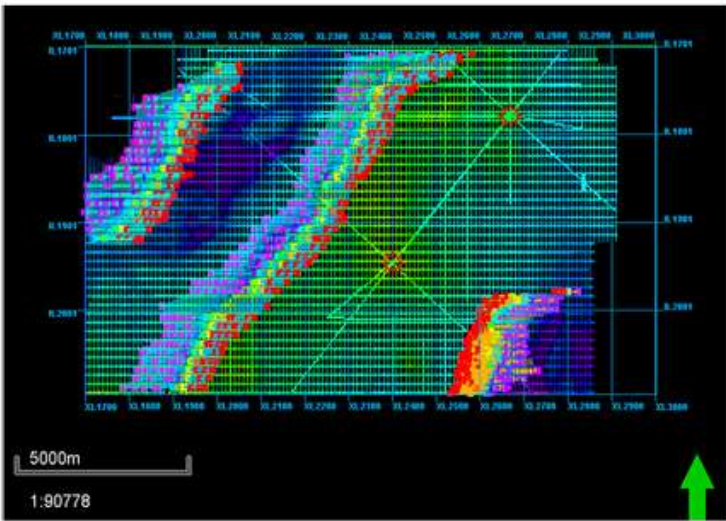
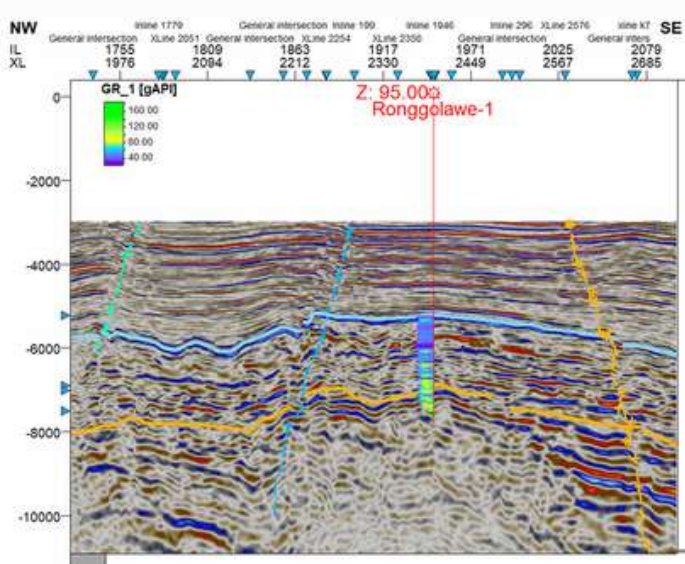
Support field development planning by improving accuracy of reservoir mapping and fault identification

Action

Performed well log & petrophysical analysis: checkshot evaluation, well-to-seismic tie, log analysis, depth surface map and attribute analysis

Results

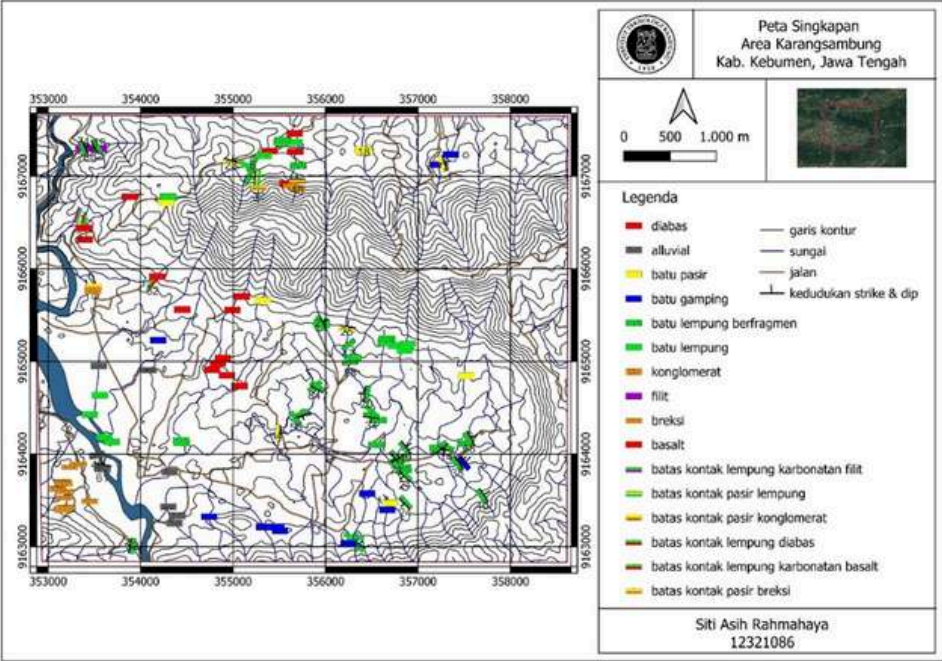
Delivered structural interpretation (horizons & faults) that clarified reservoir distribution, Gross Rock Volume Calculation, and provided key inputs for subsurface team’s development strategy



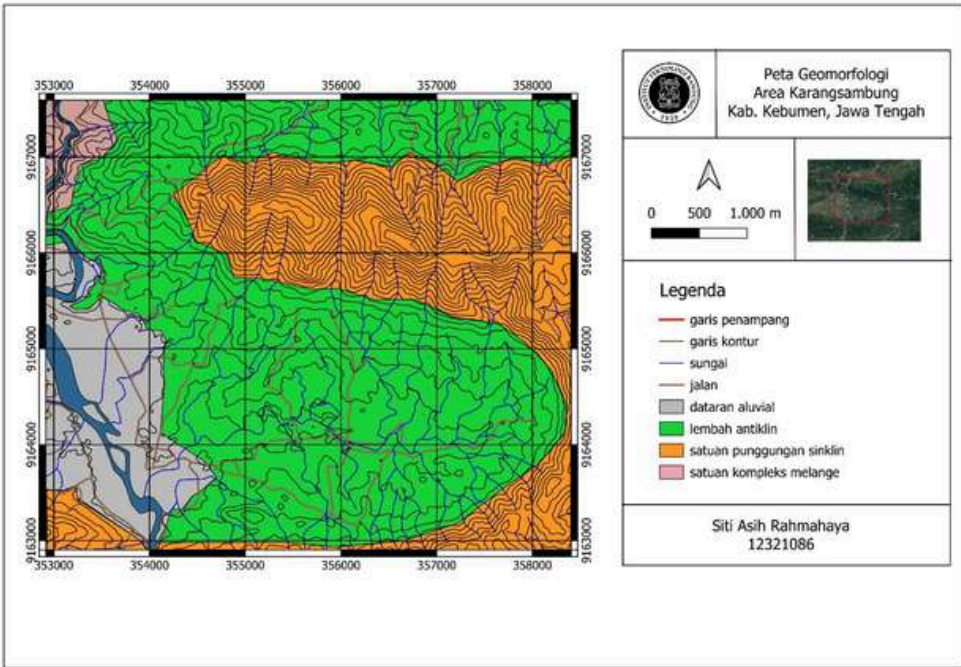
Fieldwork Experience

Fieldwork Course of Geology & Geophysics Method at Karangsembung, Kebumen, Central Java

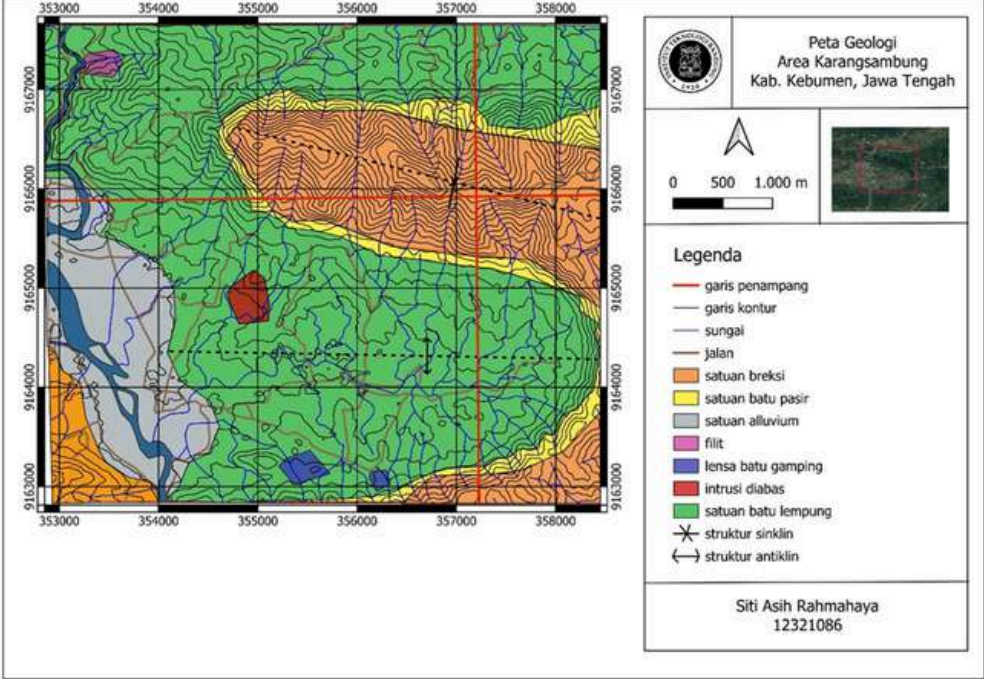
Participated in a 15-day integrated geological and geophysical field course focusing on subsurface mapping and interpretation in Karangsembung, Central Java. Conducted geological mapping (lithology identification, geomorphology, stratigraphy, and structural analysis) and acquired geophysical data using gravity, magnetic, geoelectric (VES and profiling), electromagnetic (EM and ADMT), seismic refraction, and HVSR methods. Processed and interpreted data to construct geological models and analyze subsurface structures and sediment characteristics. Strengthened teamwork, field data acquisition, analytical thinking, and report preparation skills.



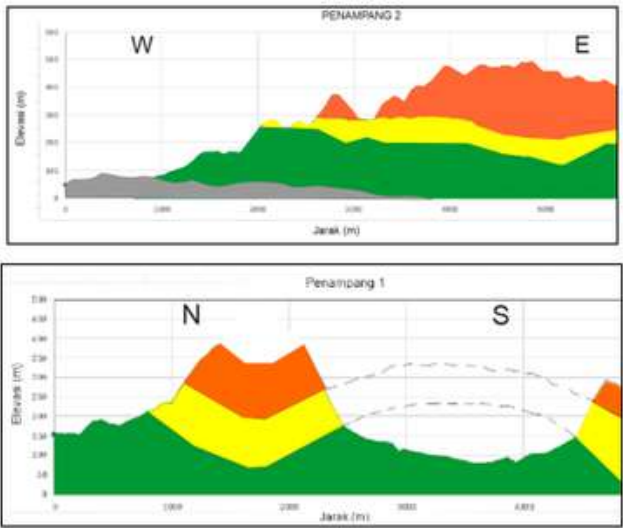
Outcrop Map



Geomorfology Map



Geological Map



Geological Cross-section Map

Fieldwork Course of Geology & Geophysics Method at Karangsembung, Central Java

Geophysics Methods: Gravity

Situation

Conducted gravity survey around BRIN area, Karangsembung, to identify subsurface density variations related to lithological and structural features.

Task

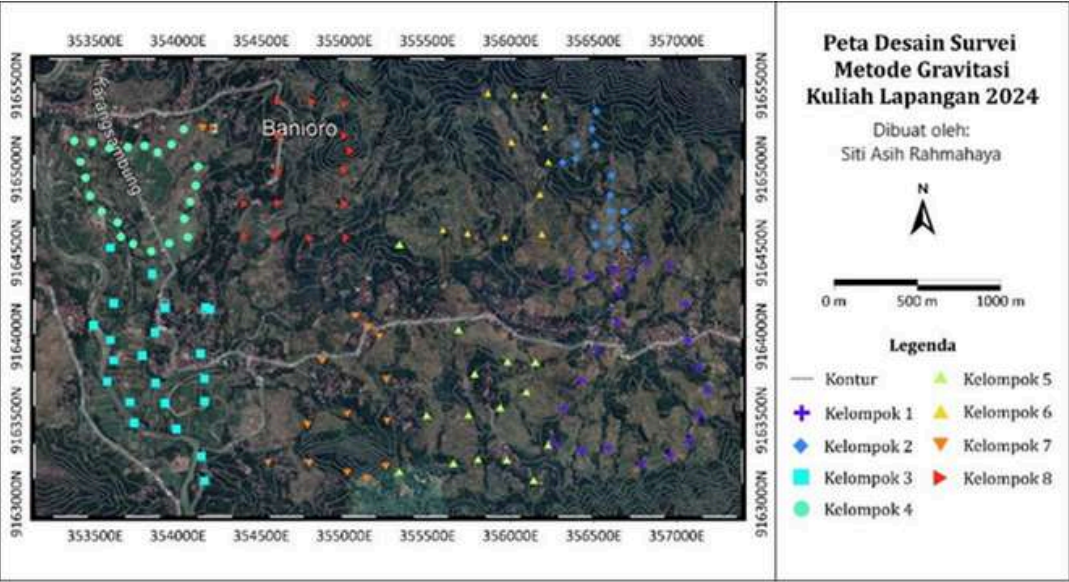
Improve understanding of subsurface structure and rock density distribution for geological interpretation.

Action

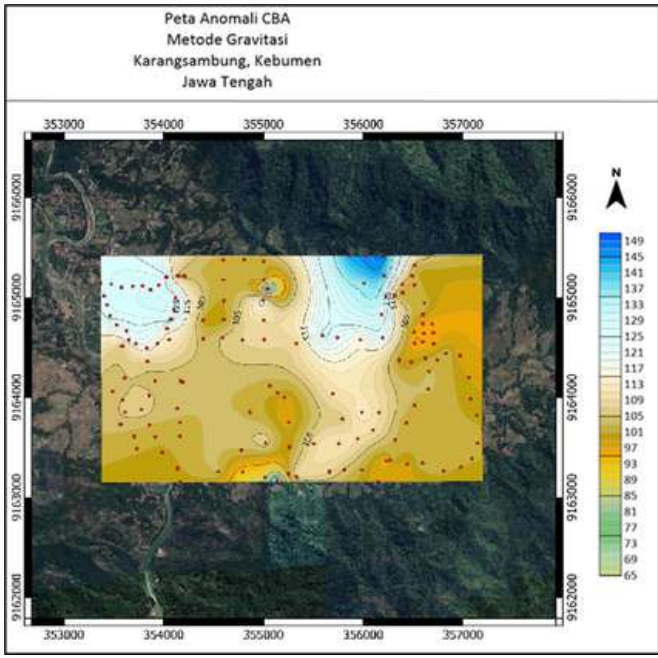
Acquired gravity data using LaCoste–Romberg gravimeter, performed data corrections (drift, latitude, Bouguer), and processed Complete Bouguer Anomaly (CBA) maps using Excel, Surfer, and ModelVision.

Results

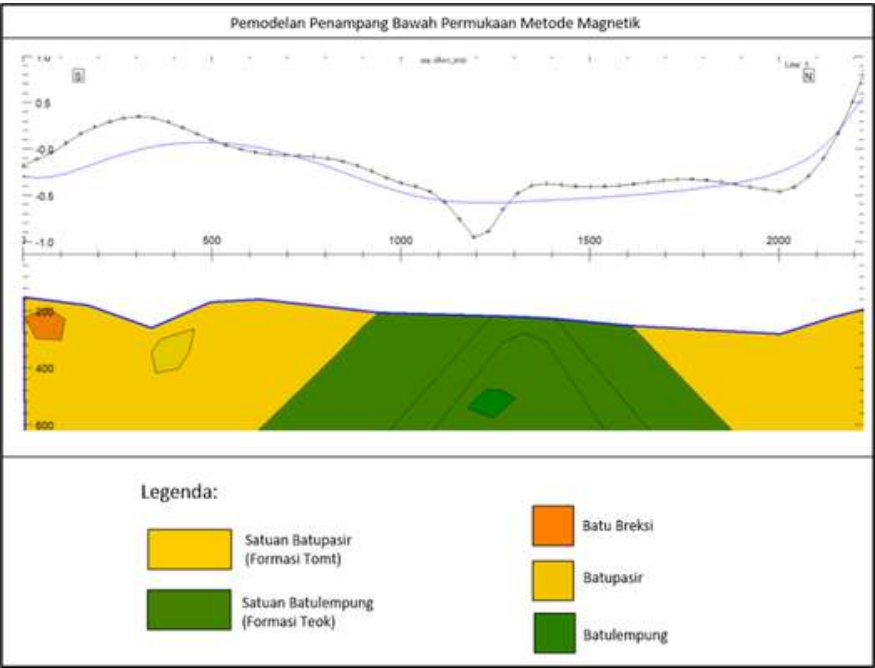
Generated Design Survey Map, CBA, regional, residual anomaly maps and gravity model that revealed subsurface high-density zones associated with basaltic and diabase intrusions, supporting structural interpretation of the area.



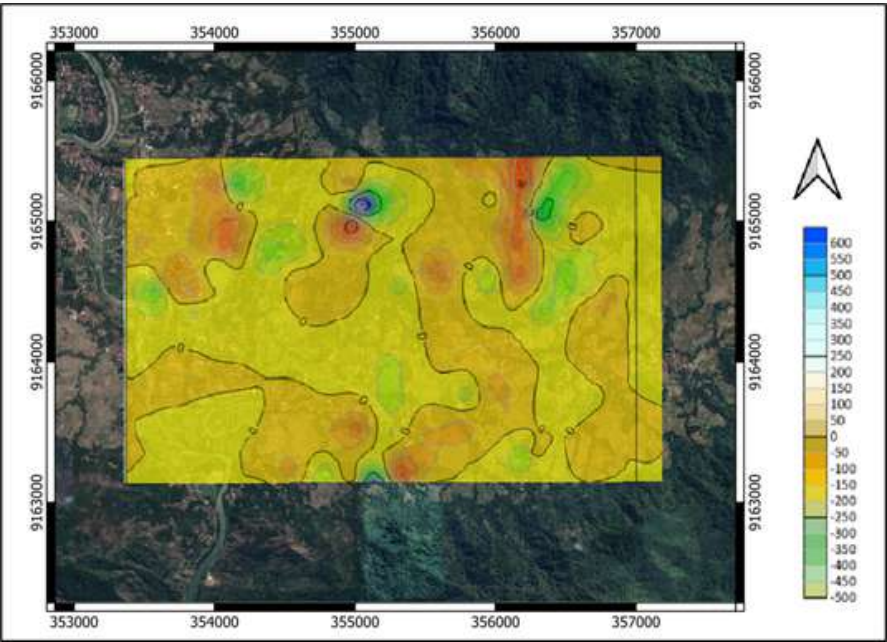
Design Survey Map



CBA Anomaly Map



Gravity Modelling



Slicing Modelling Area of Residual CBA Map

Fieldwork Course of Geology & Geophysics Method at Karangsembung, Central Java

Geophysics Methods: Magnetic

Situation

Executed magnetic survey in the BRIN area to detect subsurface magnetic anomalies and infer lithological variations.

Task

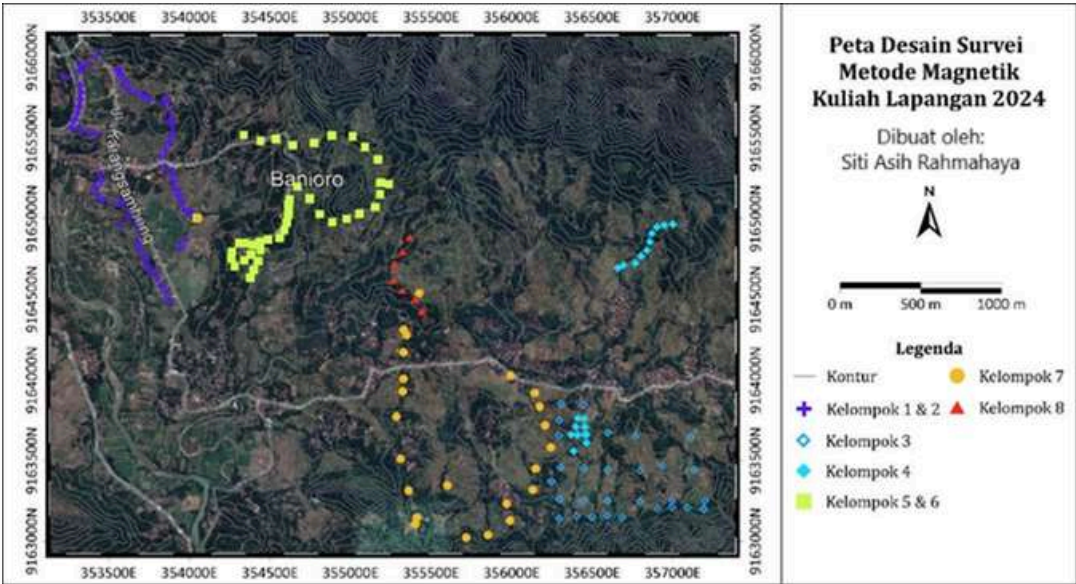
Enhance geological mapping accuracy through subsurface susceptibility contrast analysis.

Action

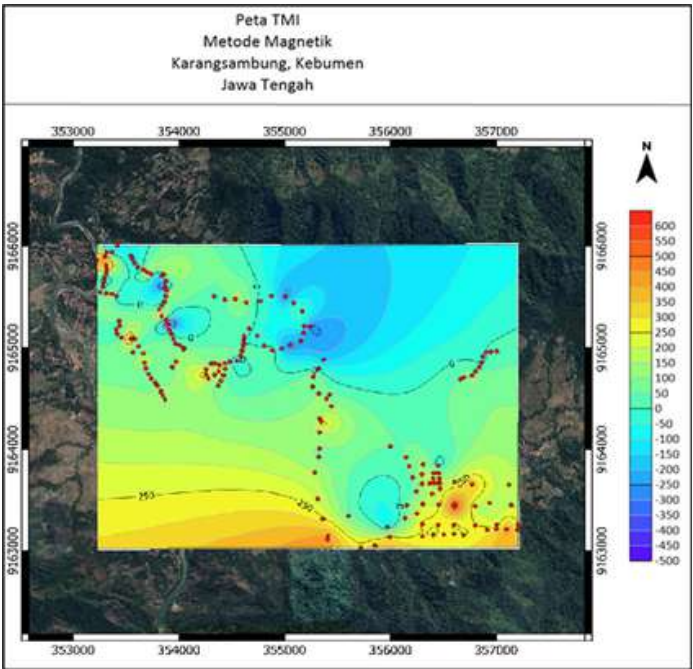
Collected magnetic data using Proton Precision Magnetometer GEM-GSM19T, applied IGRF and diurnal corrections, and performed forward modeling using Oasis Montaj and ModelVision.

Results

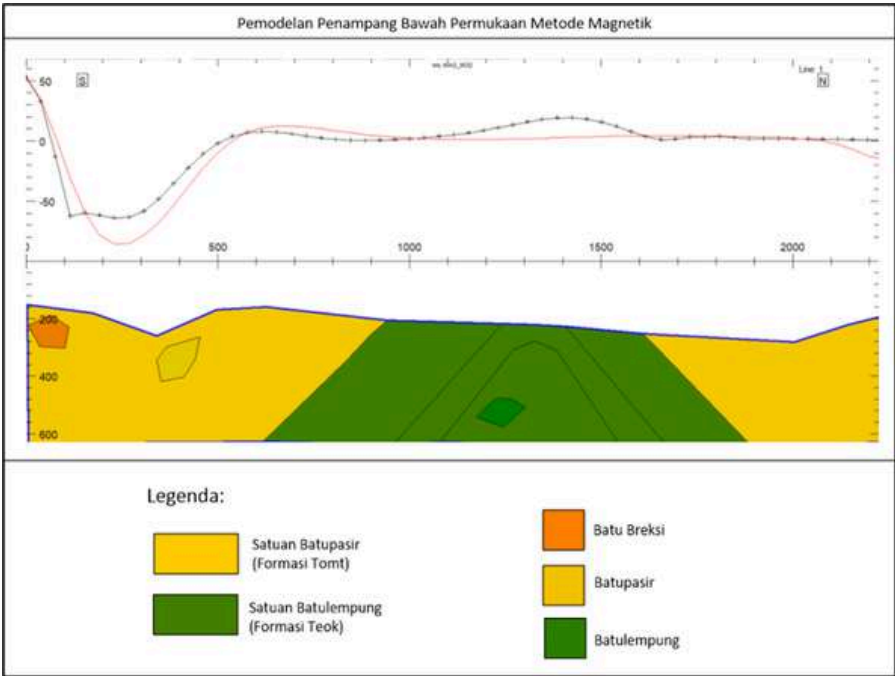
Produced Total Magnetic Intensity (TMI) maps highlighting high-susceptibility basaltic formations, aiding the identification of Waturanda Formation and associated structures.



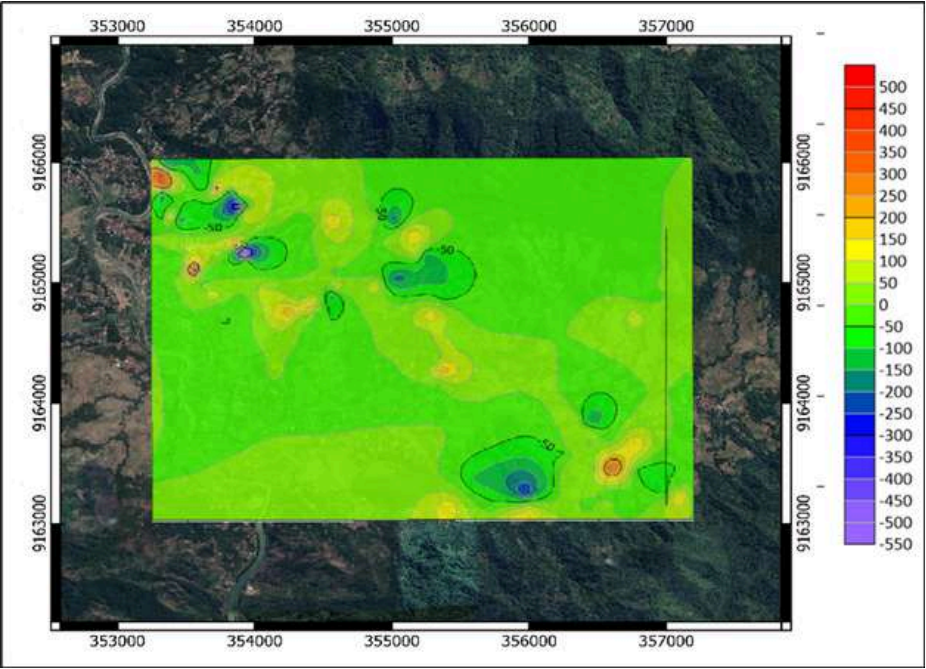
Design Survey Map



TMI Map



Magnetic Modelling



Slicing Modelling Area of Residual TMI Map

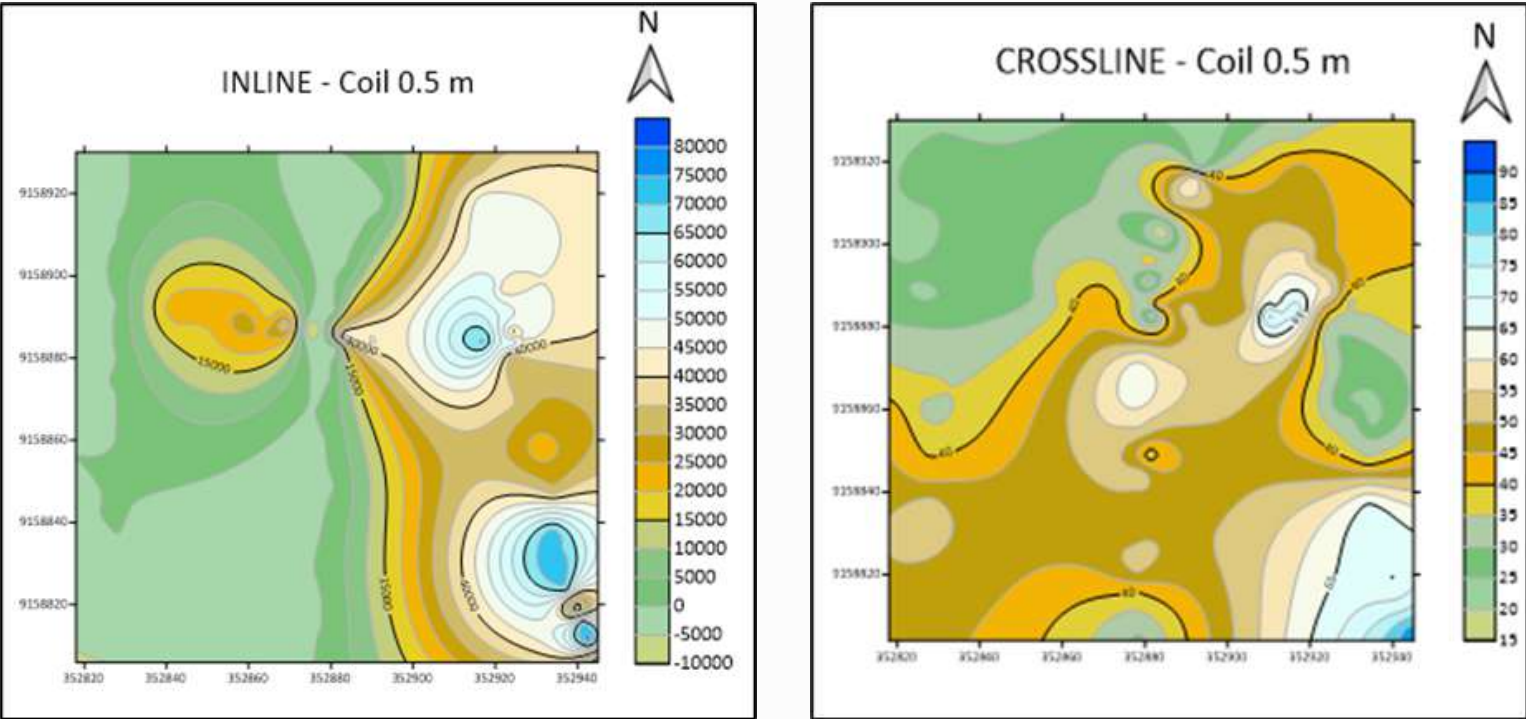
Fieldwork Course of Geology & Geophysics Method at Karangsembung, Central Java

Geophysics Methods:

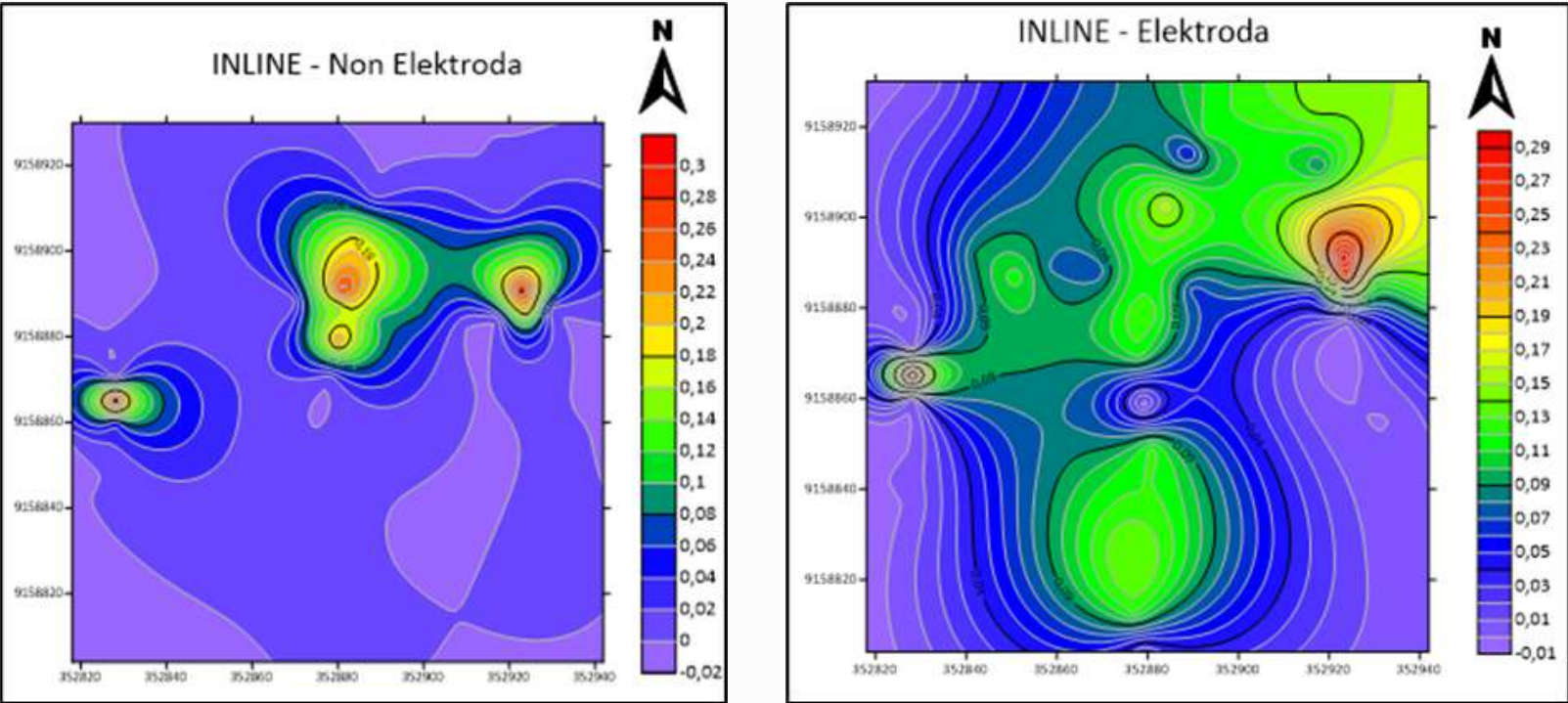
Electromagnetic

ADMT

Situation	Implemented EM38-MK2 survey in Seling area to map near-surface conductivity variations.	Carried out frequency-domain electromagnetic (ADMT) survey to investigate subsurface resistivity distribution.
Task	Support lithological differentiation and soil property analysis based on conductivity contrast.	Compare ADMT response with other electrical methods to validate near-surface structure interpretation.
Action	Recorded real-time EM data using EM38MK2xp software, converted data (N38 to M38), and processed using Surfer for contour visualization.	Used ADMT-800HT2 instrument in inline and crossline configurations (electrode and non-electrode modes); processed data using AIDU Exploration and Surfer.
Results	Produced conductivity maps that identified silty-clayey zones and correlated with ADMT and geoelectric anomalies for integrated interpretation.	Generated resistivity contour maps delineating conductive clay zones and verified consistency with geoelectric results.



Results of Electromagnetic Processing

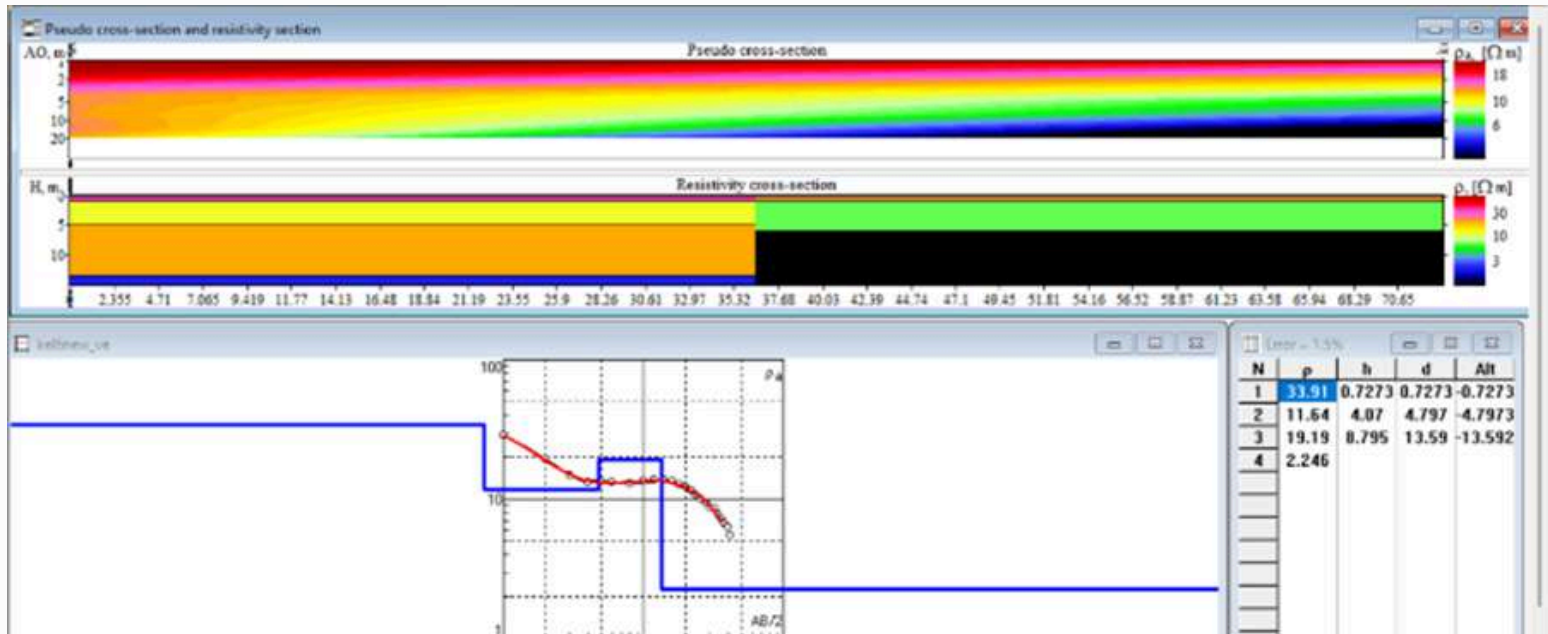


Results of ADMT Processing

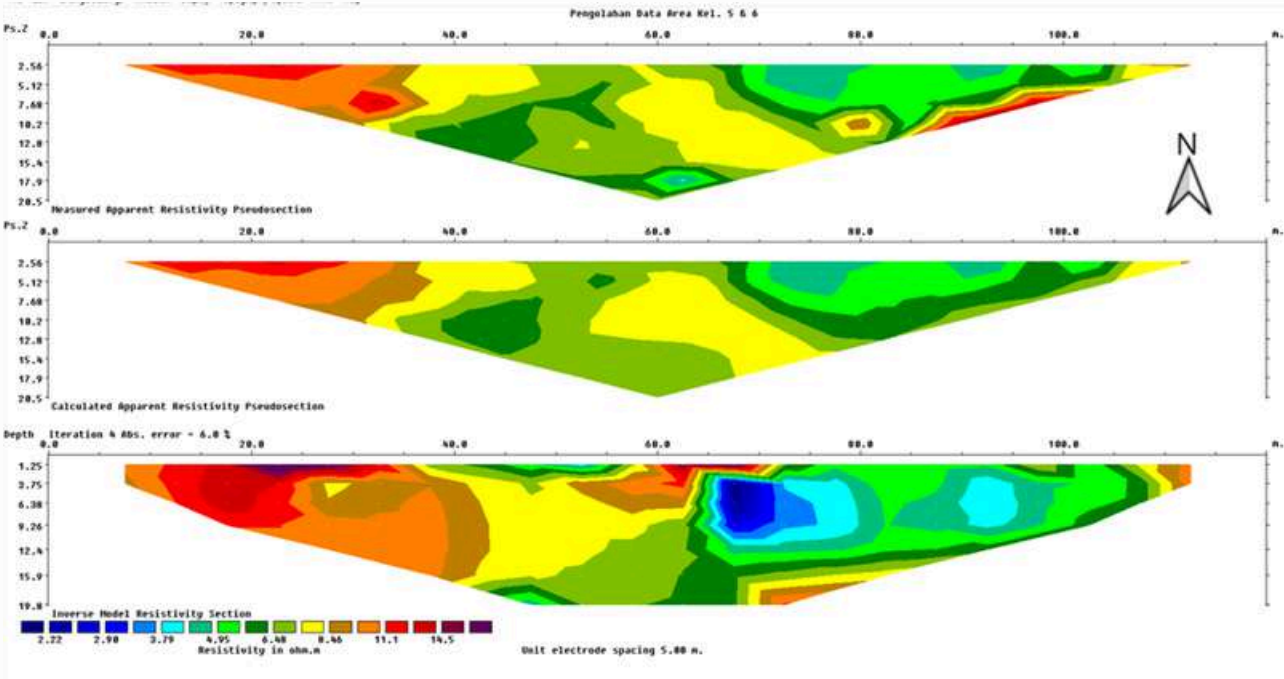
Fieldwork Course of Geology & Geophysics Method at Karangsembung, Central Java

Geophysics Methods: Sounding Profiling

Situation	Conducted Vertical Electrical Sounding (VES) survey in Seling Village to characterize vertical resistivity variations and stratigraphic layering.	Performed geoelectric profiling to map lateral resistivity variations in the Seling area.
Task	Determine subsurface lithology and correlate resistivity with borehole data for stratigraphic interpretation.	Identify horizontal lithological changes and potential groundwater zones.
Action	Acquired data using OYO McOHM EL 3i with Schlumberger configuration; processed data in IP2Win and Excel for layer resistivity modeling.	Acquired data using Mini-Sting R1 with Wenner configuration; processed results in Res2Dinv and Surfer for 2D resistivity sections.
Results	Identified four resistivity layers corresponding to soil, clay, and sand formations up to ~10 m depth, consistent with core data from Well-02.	Produced 2D resistivity profiles showing transitions from clayey-silt to sandy-clay zones and detected low-resistivity anomalies potentially indicating groundwater presence.



Pseudo Cross Section, Resistivity Cross Section & Inversion Graphic of Sounding Processing Results



Results of Geoelectric profiling Processing

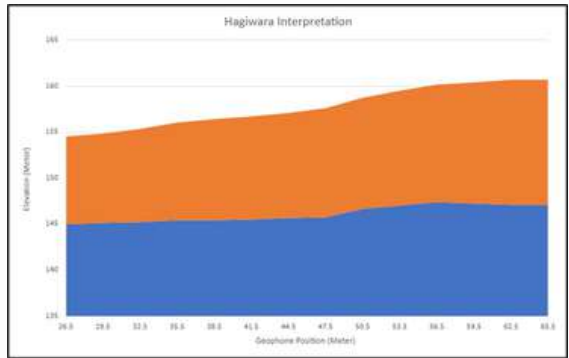
Fieldwork Course of Geology & Geophysics Method at Karangsembung, Central Java

Geophysics Methods:

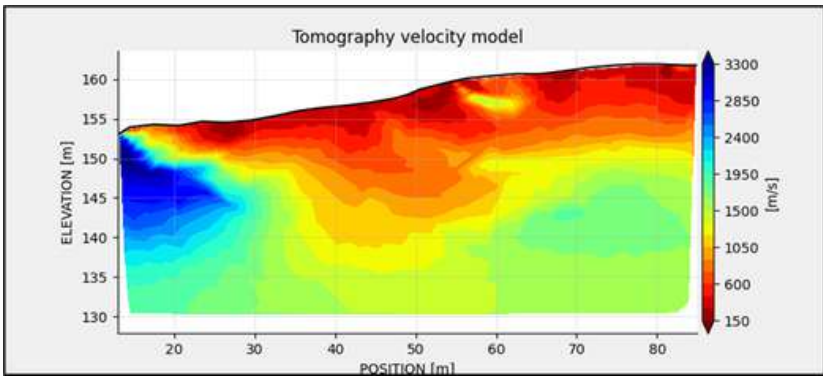
Seismic Refraction

HVSR

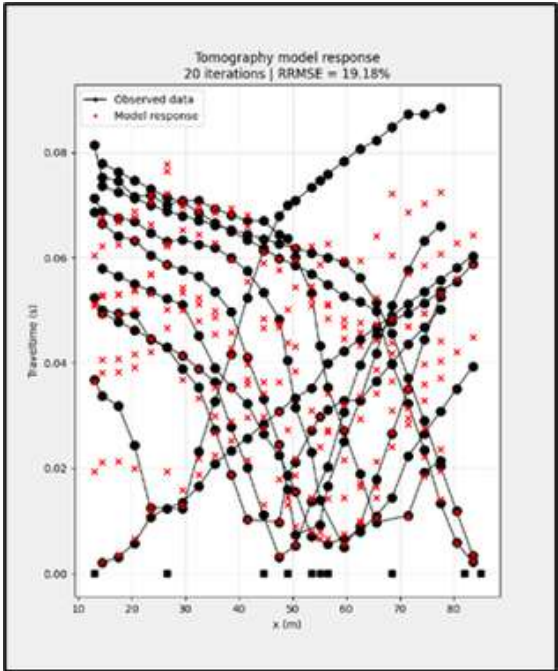
Situation	Conducted seismic refraction survey to determine subsurface velocity structure and layer thickness.	Performed HVSR measurements to analyze site response characteristics and sediment thickness.
Task	Estimate soil and rock properties and validate subsurface layering with well data.	Evaluate ground amplification and resonance frequency for seismic vulnerability assessment.
Action	Acquired data using OYO MacSeis with 24 geophones; processed data in Excel and Refrapy for Hagiwara and tomographic modeling.	Acquired microtremor data at 15 stations; processed using Geopsy and QGIS to generate amplification (A_0) and dominant frequency (f_0) maps.
Results	Derived P-wave velocity models indicating two layers: loose soil (~300 m/s) and clayey-sand (~2000 m/s), consistent with core and resistivity data.	Mapped amplification zones ($A_0 = 3-6$) and dominant frequencies (2-34 Hz), identifying moderate-to-high seismic susceptibility areas and validating subsurface sediment distribution.



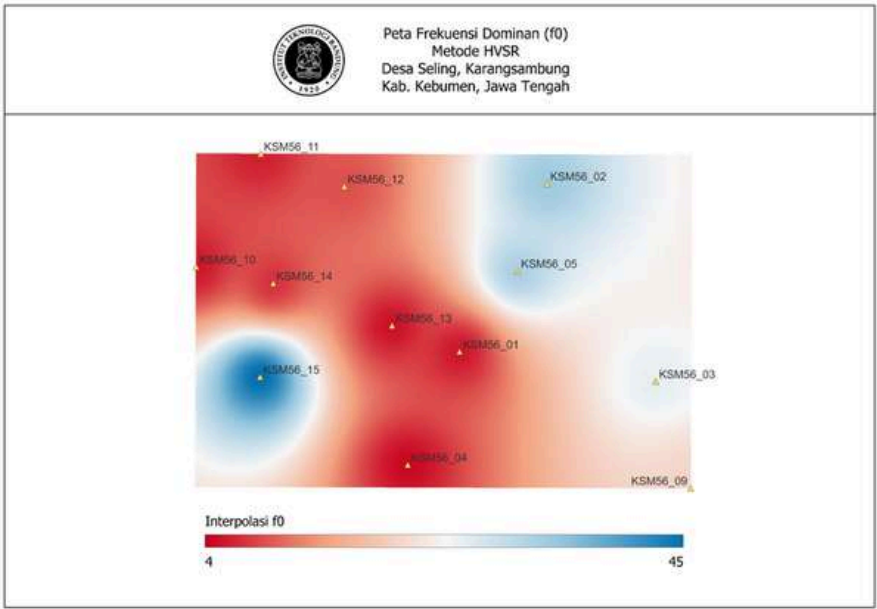
Hagiwara Cross Section



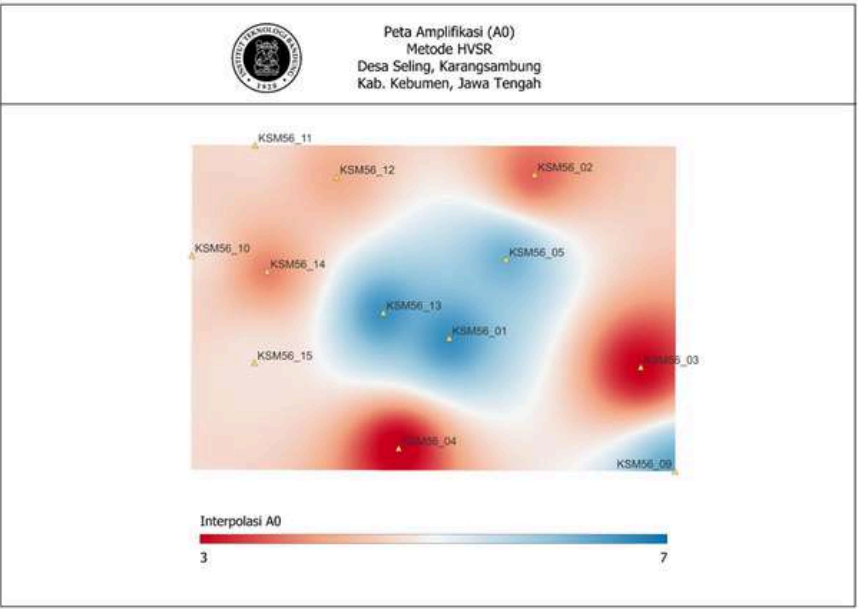
Tomography Velocity Model



Tomography Model
with error 19.18%



Frequency Dominant (F0) Map



Amplification (A0) Map

ORGANIZATIONAL & LEADERSHIP EXPERIENCES

Society of Exploration Geophysicists (SEG) ITB Student Chapter – Head of Professionalism Division

July 2023 – Jan 2024

- Led a team of 5 members to design and deliver 2 technical development programs (including SEG Focus Program and SEG Your Signal)
- Overseeing the organization and execution of activities aimed at developing members' skills and knowledge regarding professionalism in Geophysics.

Geophysics Engineering Student Association 'TERRA' ITB Chief of October Graduation Ceremony of Geophysics Student Association 'TERRA' ITB 2023

Aug 2023 – Oct 2023

- Organize all activities related to managerial activity towards the entire process of October Graduation Ceremony of Geophysics Student Association 'TERRA' ITB 2023.
- Facilitate several coordination with the main committee of October Graduation Ceremony and the entire graduate of Geophysics Student of October 2023.

Society of Exploration Geophysicists (SEG) ITB Student Chapter – Head of Professionalism Division

July 2023 – Jan 2024

- Led a team of 5 members to design and deliver 2 technical development programs (including SEG Focus Program and SEG Your Signal)
- Overseeing the organization and execution of activities aimed at developing members' skills and knowledge regarding professionalism in Geophysics.



Student Development
Departement



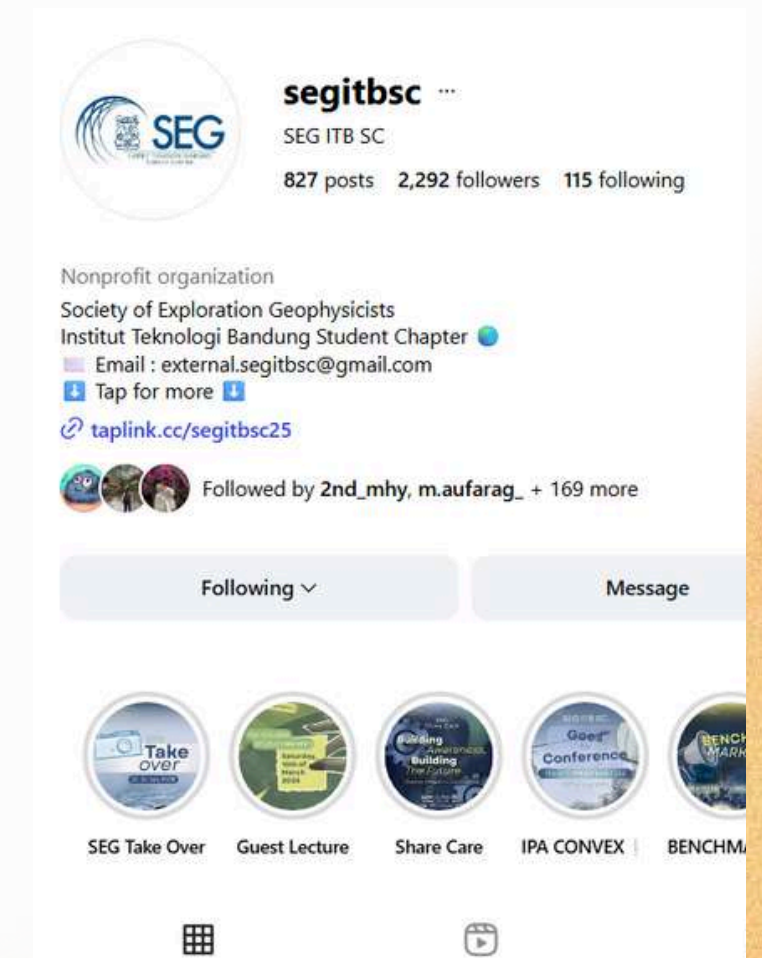
SEG Your Signal 1.0



SEG Your Signal 2.0



SEG Focus



Instagram of @segitbsc

Geophysics Engineering Student Association 'TERRA' ITB

Head of October Graduation Ceremony of Geophysics Student Association 'TERRA' ITB 2023

Aug 2023 – Oct 2023

- Responsible for all activities related to managerial activity towards the entire process of October Graduation Ceremony of Geophysics Student Association 'TERRA' ITB 2023, including WisDay, Arak-Arakan, Wisnight, Arak-arakan sore, and Instagram Takeover
- Facilitate all coordination with the main committee of October Graduation Ceremony and the entire graduate of Geophysics Student of October 2023.



WisDay



Arak-arakan



Arak-arakan Sore



Wisnight



IG Takeover



Committee Structure

ACHIEVEMENTS

- 1st winner of **Paper Poster Competition** 'Geoscience and Competition GEOTRAP' Padjajaran University, 2023.

Paper title: Characterization of the Mount Talang Geothermal System, West Sumatra, Based on the Integration of Geological, Geochemical, and Geophysical Data.

- **Dean's List Award**, Faculty of Mining and Petroleum Engineering, Institut Teknologi Bandung — Semester II 2023/2024

- Finalist of **Paper Poster Competition** 'Petroleum Integrated Days (PETROLIDA)' Sepuluh November Institute of Technology, 2024.

Paper title: Optimizing Oil Buffer Reserve Using Petrophysical Analysis and Seismic Interpretation in The Global Energy Security Initiatives.

- 2nd runnerup **Paper Poster Competition** 'International Undergraduate Geophysics Competition (IUGC)' Bandung Institute of Technology, 2024.

Paper title: Characterization of Geothermal Energy Potential at Gedong Songo, Mount Ungaran, Central Java: The Hydrogeochemistry Study to Support Net-Zero Emissions Initiatives

[Raih 2nd Runner-Up di IUGC ITB 2024, Mahasiswa Teknik Geofisika ITB Bawakan Analisis Potensi Energi Geotermal Gedong Songo - Institut Teknologi Bandung](#)

- 1st Winner of **Interpretation Geophysics Data** Competition, Mulawarman University, 2024.

[Tim Gassin ITB Juara 1 Lomba Interpretasi Data Seismik HMGF FMIPA Universitas Mulawarman - Institut Teknologi Bandung](#)

- 1st Winner Lomba Karya Tulis Ilmiah Nasional (LKTIN) 'Geodetic Expo' Diponegoro University, 2024.

Paper title: Seismicity Study Based on United States Geological Survey (USGS) Repository Data Over a Five-Year Period (2019–2023) as an Effort to Mitigate Earthquake Disasters in Indonesia.

[Tim Mahasiswa Teknik Geofisika ITB Juara 1 LKTIN Geodetic Expo Vol. I Universitas Diponegoro - Institut Teknologi Bandung](#)

- 2nd runner up of **Tes Wawasan Kebumian** 'Aceh Youth Geoscience Activity' Syiah Kuala University, 2024.

[Tim Gassin ITB Juara 3 Tes Wawasan Kebumian Aceh Youth Geoscience Activity 2024 - Institut Teknologi Bandung](#)

[Tim Gassin ITB Juara 3 Tes Wawasan Kebumian Aceh Youth Geoscience Activity 2024 - \[www.breakingnewsbandung.com\]\(http://www.breakingnewsbandung.com\)](#)

- 1st runner up **Paper Poster Competition** of 'Geophysics Festival (GEOFEST)', Sumatera Institute of Technology, 2024.

Paper title: Spatial Analysis of b-Value from 2019 to 2023 for Seismic Hazard Mitigation in The Indonesian Region

- 3rd runner up of **Case Study Competition** of SUPERBOWL, Bandung Institute of Technology Society of Petroleum Engineer, 2024.

ACHIEVEMENTS



2nd runnerup Paper Poster Competition 'International Undergraduate Geophysics Competition (IUGC)'



Finalist of Paper Poster Competition 'Petroleum Integrated Days (PETROLIDA)'



1st Winner of Interpretation Geophysics Data Competition



1st runner up Paper Poster Competition of 'Geophysics Festival (GEOFEST)'



3rd runner up of Case Study Competition of SUPERBOWL



1st Winner Lomba Karya Tulis Ilmiah Nasional (LKTIN) 'Geodetic Expo' Diponegoro University



2nd runner up of Tes Wawasan Kebumian 'Aceh Youth Geoscience Activity'

OTHER CAMPUS ACTIVITIES



Operational Division's Secretary of Olimpiade KM ITB



Leader of SEG Women's Network Community



Management Personalia (MSDA) Staff of HIMA TG "TERRA" ITB



Mentor of OSKM ITB 2022



Staff of Organizational Research & Development Division - MWA WM ITB



Vice Head Operations of Kaderisasi Wilayah FTTM ITB 2022



SEG ITB SC's Company visit to Baker Hughes

THANK YOU! LET'S CONNECT!

Linkedin

www.linkedin.com/in/sitiasihrahmahaya

Email Address

sitiasihrahmahaya@gmail.com

Phone Number

[083116389496](tel:083116389496)

Instagram

[@asihrhmhaya](https://www.instagram.com/asihrhmhaya)

Github

[Github/Project](#)